

Statistical Analysis of Spotify Song Popularity

Project Proposal and Analysis Plan

Group 3

MATH 573 – Statistical Consulting

September 2026

What makes a Spotify song successful?

We will study the musical, temporal, artist, and cross-platform characteristics associated with streaming success from 2023 to 2025.

Four Groups of Factors

We look at four groups of factors:

- **Music** – audio features like energy, danceability, tempo
- **Artist** – solo versus collaboration, artist popularity
- **Platforms** – Spotify, YouTube, TikTok, Apple Music, Deezer, Shazam
- **Time** – release year, release month, song age

All four are examined as potential drivers of **streaming success**.

Why This Is More Than a Spotify Dashboard

We will move from

- descriptive statistics
- visual exploration
- correlations
- statistical tests

To

- regression modeling
- multivariate analysis
- clustering
- prediction and model comparison

Goal: explain patterns statistically, not simply display them.

Spotify 2023 to 2025

- streaming counts
- playlists and charts
- release information
- audio characteristics
- artist information

Additional platform data

- Apple Music
- Deezer
- YouTube
- TikTok
- Shazam
- other available platform measures

Key methodological issue: the datasets do not have identical schemas.

Data Integration Strategy

Step 1. Collect the 2023, 2024, 2025, and Wrapped 2025 datasets.

Step 2. Harmonize the schemas and identify comparable variables across years.

Step 3. Build two analytical datasets:

- a **harmonized longitudinal dataset** (comparable across years)
- a richer **platform-specific dataset** (cross-platform analysis)

Only variables that are sufficiently comparable and statistically defensible will be carried forward.

Core Variables

Popularity

- Spotify streams
- playlists
- charts
- cross-platform metrics

Musical

- BPM
- danceability
- energy
- valence
- acousticness
- instrumentality

Context

- release year
- release month
- artist count
- year of dataset
- song age

What characteristics are associated with Spotify streaming success?

- ① How does streaming success differ across 2023, 2024, and 2025?
- ② Which musical characteristics are associated with higher streaming counts?
- ③ Does song age relate to streaming success?
- ④ Do collaborations perform differently from solo songs?
- ⑤ How concentrated is streaming success among artists?
- ⑥ How strongly does Spotify popularity correspond to other platforms?
- ⑦ Can song and platform characteristics explain variation in streams?
- ⑧ Can songs be grouped into distinct musical and popularity profiles?
- ⑨ Can we classify songs as high- or lower-streaming?

From Questions to Methods

Research question	Planned method
How are the datasets distributed?	Descriptive statistics and EDA
Do years differ?	ANOVA and Kruskal-Wallis
Which variables are associated?	Pearson and Spearman correlation
Which factors explain streams?	Multiple regression
Are effects robust?	Diagnostics and robust standard errors
Are there song profiles?	PCA and clustering
Can popularity be classified?	Logistic regression and ML
How concentrated is success?	Gini and Pareto analysis

Step 1 – Acquire and Inspect the Data

The workflow:

Kaggle sources → Programmatic download → Schema inspection → Data-quality report

Questions at this stage

- What does one row represent?
- What variables are available in each year?
- Which variables are numeric, categorical, or dates?
- Which variables can legitimately be compared across years?

Step 2 – Clean the Data

We will address, in order:

- Missing values
- Duplicates
- Data types
- Invalid values
- Outliers
- Consistency
- Documentation

Principle: an extreme value is not automatically an error.

Step 3 – Feature Engineering

We will construct variables that have direct statistical meaning.

Song age

$$\text{Song Age} = \text{Dataset Year} - \text{Release Year}$$

Streaming transformation (streams are right-skewed):

$$Y = \log(1 + \text{Streams})$$

Collaboration indicator:

$$C_i = 1 \text{ if artist count} > 1, \quad C_i = 0 \text{ otherwise}$$

What Does a Typical Hit Look Like?

We will examine:

- mean, median, variance, standard deviation
- quartiles and interquartile range
- skewness and distribution shape
- extreme observations
- differences between raw and log-stream distributions

A key issue to investigate is whether a small number of extreme hits dominate the mean.

Did the distribution of streaming success change over time?

We will compare the three yearly means using both parametric and nonparametric approaches.

Planned analysis

- boxplots and distribution plots
- one-way ANOVA
- Kruskal-Wallis test
- post-hoc comparisons when appropriate

Are Musical Features Associated With Streams?

We will examine relationships between **streams** and energy, danceability, valence, BPM, acousticness, and similar features.

Pearson correlation:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Spearman correlation is the correlation of ranks.

Correlation does not imply causation.

Hypothesis Testing

Year effect

$$H_0 : \mu_{2023} = \mu_{2024} = \mu_{2025}$$

Collaboration effect

$$H_0 : \mu_{\text{solo}} = \mu_{\text{collab}}$$

Key and popularity

$$H_0 : \text{key and popularity are independent}$$

We will report: test statistic, p -value, effect size, and 95% CI.

Statistical Rigor

Because many variables will be investigated, we will address:

- multiple hypothesis testing
- effect size versus statistical significance
- assumptions of statistical tests
- nonparametric alternatives
- confidence intervals

For multiple testing, we may use Bonferroni or false-discovery-rate control.

$$\alpha^* = \frac{\alpha}{m}$$

for a Bonferroni adjustment.

Can We Explain Streaming Success?

Our central regression model will use log streams:

$$\log(1 + S_i) = \beta_0 + \beta_1 D_i + \beta_2 E_i + \beta_3 V_i + \cdots + \epsilon_i$$

Progressively richer models

- **Model 1:** musical features
- **Model 2:** plus song characteristics
- **Model 3:** plus year
- **Model 4:** plus platform exposure

Models will be compared using fit, parsimony, diagnostics, and predictive performance.

Regression Diagnostics

We will not treat a regression output as the final answer.

We will investigate:

- multicollinearity
- residual patterns
- heteroskedasticity
- influential observations
- model specification
- robust standard errors

Example diagnostic question:

$$H_0 : \text{Var}(\epsilon_i) = \sigma^2$$

versus heteroskedasticity.

PCA: Can We Reduce the Audio Dimensions?

Many audio variables may contain overlapping information. PCA constructs:

$$Z_1 = a_{11}X_1 + \cdots + a_{1p}X_p$$

such that the first component explains the maximum possible variance.

Questions

- How many dimensions are needed?
- Which audio variables drive each component?
- Is there a latent musical structure?

From many correlated variables to a smaller set of interpretable dimensions.

Clustering: Are There Types of Hit Songs?

After standardization, we will investigate whether songs form natural groups.

$$\min_{\{C_k\}} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

We will determine the number of clusters using evidence

- silhouette score
- within-cluster variation
- cluster stability
- interpretability

Not four clusters because four looks good, but four clusters because the data support them.

Is Spotify Success Universal?

For songs with cross-platform data, we will compare Spotify with YouTube, TikTok, Apple Music, Deezer, and Shazam.

Does popularity on one platform translate to popularity on another?

We will use correlation and multivariable models to identify which platform measures are most strongly associated with Spotify streams.

Is Streaming Success Concentrated?

We will examine whether a small number of artists or songs account for a disproportionate share of streams.

Tools

- artist-level aggregation
- Pareto analysis
- concentration curves
- Gini coefficient

The Gini coefficient is:

$$G = \frac{\sum_i \sum_j |x_i - x_j|}{2n \sum_i x_i}.$$

Can We Identify a High-Streaming Song?

Define:

$$Y_i = 1 \text{ if } S_i \geq Q_{0.75}, \quad Y_i = 0 \text{ if } S_i < Q_{0.75}$$

where $Q_{0.75}$ is the 75th percentile of streams.

Logistic regression:

$$\log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik}.$$

We will interpret coefficients using odds ratios:

$$OR_j = e^{\beta_j}.$$

Prediction Is Not the Same as Explanation

We will separate:

- **Explanation** – statistical inference
- **Prediction** – out-of-sample performance

Predictive models will be evaluated on held-out data using:

accuracy, precision, recall, F_1 , ROC-AUC.

11-Week Project Plan

Weeks	Planned work
1–2	Data acquisition, schema inspection, cleaning, harmonization
3	Feature engineering and master analytical datasets
4	Descriptive statistics and exploratory data analysis
5	Temporal, artist, and release analyses
6	Correlation analysis and hypothesis testing
7–8	Multiple regression, diagnostics, and robust inference
9	Cross-platform analysis and artist concentration
10	PCA, clustering, and cluster validation
11	Logistic regression, prediction, and final integration

Deliverables by Phase

Phase	Weeks	Deliverable
Phase I: Preparation	1–3	Clean, harmonized, engineered datasets
Phase II: Exploration	4–5	EDA report and temporal/artist trends
Phase III: Inference	6–8	Correlation, hypothesis tests, regression
Phase IV: Multivariate	9–10	Cross-platform, concentration, PCA, clusters
Phase V: Prediction	11	Classifier, final report, presentation

Each phase produces an auditable result before the next phase begins.

How We Will Work as a Group

Shared responsibilities

- agree on research questions
- maintain one data dictionary
- use reproducible code
- document every preprocessing decision
- review each other's analyses

Reproducibility

- programmatic data acquisition
- fixed random seeds where needed
- version-controlled analysis
- saved cleaned datasets
- documented statistical decisions

No analysis without documenting why the method was chosen.

What We Hope to Learn

- ➊ **Musical characteristics** – which audio features are associated with streaming success?
- ➋ **Time** – has the profile of popular songs changed from 2023 to 2025?
- ➌ **Platforms** – is popularity transferable across digital platforms?
- ➍ **Artists** – how concentrated is streaming success?
- ➎ **Statistical structure** – are there identifiable groups of songs?
- ➏ **Prediction** – how accurately can observable characteristics identify high-streaming songs?

Limitations We Will Address

- Kaggle datasets may represent selected popular songs rather than all Spotify songs.
- Dataset schemas differ across years.
- Cross-platform variables are not available for every dataset.
- Observational data support association, not automatic causal claims.
- Streaming counts may reflect exposure, promotion, and artist popularity.
- Differences in data collection may affect comparisons across years.

These limitations will be treated as part of the statistical analysis.

From Raw Data to Statistical Conclusions

**Raw data → Clean → Harmonize → Engineer → Explore → Infer → Model → Validate
→ Interpret → Communicate**

The objective is not to find the most impressive graph.

The objective is to answer meaningful questions with defensible statistics.

The Project in One Sentence

We will investigate how musical, temporal, artist, and cross-platform characteristics are associated with Spotify streaming success using a reproducible statistical workflow from data cleaning through inference, modeling, clustering, and prediction.

Thank You

Questions and suggestions